Learning Objective 1.3: I can describe signal-space concepts (basis functions, constellations, decision regions) and apply them to a digital modulation scheme.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Constellations and data rate.** A link must carry 1 Mb/s.

   (a) How many bits per symbol do 8-PSK and 64-QAM carry?

   $k = \log_2 M: \quad \log_2 8 = 3, \quad \log_2 64 = 6$

   8-PSK: 3 bits

   64-QAM: 6 bits

   (b) Find the symbol rate the link needs with QPSK and with 16-QAM.

   $$R_s = \frac{R_b}{k}$$

   QPSK: $R_s = \frac{10^6}{2} = 500$ ksym/s

   16-QAM: $R_s = \frac{10^6}{4} = 250$ ksym/s

   QPSK: 500 ksym/s

   16-QAM: 250 ksym/s

   (c) Sketch QPSK and 4-PAM at the same average symbol energy, and explain in one or two sentences why a radio uses QPSK rather than 4-PAM on a carrier.

   QPSK puts one point in each quadrant; 4-PAM puts four points on the I axis. Both carry two bits per symbol, but QPSK uses both basis functions and so spaces its points 1.41 apart at unit energy against 0.89 for 4-PAM. Wider spacing tolerates more noise for the same power.

2. **Noise and decisions.** A BPSK receiver sees its two points at $+1$ V and -1 V on the I axis. The noise on each axis is Gaussian with a standard deviation of 0.4 V. Use $Q(2.5) = 6.2 \times 10^{-3}$ and $Q(4) = 3.2 \times 10^{-5}$.

   (a) State the decision rule and draw the decision regions.

   Decide $+1$ if $I > 0$ and -1 if $I < 0$. The boundary is the Q axis, halfway between the two points, and the two regions are the right and left half-planes.

   (b) Find the bit error rate.

   An error needs the noise to push the sample past 0, a distance of 1 V, which is $1/0.4 = 2.5$ standard deviations:

   $$P_b = Q(2.5) = 6.2 \times 10^{-3}$$

   P_b = 6.2×10^{-3}

   (c) The noise drops to 0.25 V. Find the new bit error rate, and the change in SNR in dB.

   $$P_b = Q\left(\frac{1}{0.25}\right) = Q(4) = 3.2 \times 10^{-5}$$

   $$\Delta\text{SNR} = 20\log_{10}\left(\frac{0.4}{0.25}\right) = 4.1 \text{ dB}$$

   P_b = 3.2×10^{-5}

   ΔSNR = 4.1 dB

   (d) A 4.1 dB improvement in SNR cut the error rate by a factor of about 200. Explain why the bit error rate changes so much faster than the SNR.

   The error rate is the tail of a Gaussian beyond the decision boundary, and that tail falls off as $e^{-x^2/2}$. Moving the boundary from 2.5 to 4 standard deviations away removes almost all of the remaining tail, so a few dB of SNR changes the error rate by orders of magnitude.

3. **Eb/N0, Es/N0, and SNR.** A receiver filtered to its symbol rate measures an SNR of 15 dB.

   (a) Find E_s/N_0, and E_b/N_0 for QPSK and for 16-QAM.

   With $B = R_s$, $E_s/N_0 = \text{SNR} = 15$ dB.

   $$\text{QPSK: } \frac{E_b}{N_0} = 15 - 10\log_{10}(2) = 12.0 \text{ dB}$$

   $$\text{16-QAM: } \frac{E_b}{N_0} = 15 - 10\log_{10}(4) = 9.0 \text{ dB}$$

QPSK:	12.0 dB
16-QAM:	9.0 dB

   (b) Use the lesson's BER curves to estimate the bit error rate of each.

   QPSK at 12.0 dB is far past the 10^{-5} point (9.6 dB); the curve gives about 10^{-8}. 16-QAM at 9.0 dB is still near 4×10^{-3}.

QPSK:	$\approx 10^{-8}$
16-QAM:	$\approx 4 \times 10^{-3}$

   (c) Explain why BER curves are plotted against E_b/N_0 rather than SNR.

   Modulations carry different numbers of bits per symbol. E_b/N_0 measures the energy spent on each bit, so it compares them on equal terms; at the same SNR, 16-QAM spreads the symbol's energy over four bits and QPSK over two.

4. **Gray coding.** An 8-PSK modem places its points every 45° around a circle.

   (a) Label the eight points with a Gray code, starting from 000 at 0°.

   Going counterclockwise: 000, 001, 011, 010, 110, 111, 101, 100. Each label differs from both of its neighbors, including 100 and 000, in one bit.

   (b) With natural binary labels instead (000, 001, 010, ..., 111), how many bits does a slip from the point labeled 011 to its neighbor 100 corrupt?

   011 and 100 differ in all three positions, so all three bits are wrong.

   3 bits

   (c) The Gray-coded modem measures a symbol error rate of 3×10^{-4}. Estimate its bit error rate.

$$P_b \approx \frac{P_s}{k} = \frac{3 \times 10^{-4}}{3} = 1 \times 10^{-4}$$

   $P_b =$ 1×10^{-4}

5. **Choosing a modulation.** A link budget predicts an SNR of 21 dB in a receiver filtered to 1 Msym/s. The link needs a bit error rate of 10^{-5} and at least 6 dB of margin. At that error rate QPSK needs $E_s/N_0 = 12.6$ dB, 8-PSK needs 17.7 dB, and 16-QAM needs 19.5 dB.

   (a) Find the margin each modulation leaves.

   QPSK: $21 - 12.6 = 8.4$ dB
   8-PSK: $21 - 17.7 = 3.3$ dB
   16-QAM: $21 - 19.5 = 1.5$ dB

   QPSK: 8.4 dB

   8-PSK: 3.3 dB

   16-QAM: 1.5 dB

   (b) Which modulation meets the requirement, and what data rate does it carry?

   Only QPSK leaves 6 dB. At 1 Msym/s and 2 bits/symbol it carries 2 Mb/s.

   QPSK, 2 Mb/s

   (c) A customer wants 4 Mb/s. One option is 16-QAM at 1 Msym/s; the other is QPSK at 2 Msym/s, which needs twice the bandwidth at the same transmit power. Compare the margin of each and recommend one.

   16-QAM at 1 Msym/s leaves 1.5 dB of margin. Doubling the bandwidth doubles the noise power, so QPSK at 2 Msym/s sees an SNR of $21 - 3.0 = 18.0$ dB and leaves $18.0 - 12.6 = 5.4$ dB. Neither meets 6 dB, but QPSK in twice the bandwidth comes much closer; it is the better choice if the bandwidth is available, and otherwise the link needs about 0.6 dB more from the budget, for example a slightly higher-gain antenna.

6. **Reading a constellation.** Four QPSK receivers each show one problem on a constellation sink. Name the cause of each.

   (a) The four points sit in the middle of their quadrants, but each is a round cloud about a third of the way to the boundaries.

   Noise. The cloud's size reads the SNR; nothing else is wrong.

   (b) Four tight clouds, each rotated 30° counterclockwise from where it belongs, standing still.

   A fixed phase offset between the transmitter and receiver oscillators.

   (c) The points are smeared into a ring of constant radius.

   A frequency offset: the phase error grows continuously, so the constellation rotates through every angle during the capture.

   (d) Explain why the constellation in part (b) would produce a very high bit error rate if the rotation reached 45°, even with no noise.

   The QPSK decision boundaries are the axes, 45° from each point. A rotation of 45° puts every point on a boundary, and anything past it puts every point in a neighbor's region, so the receiver must correct the phase before it decides.

Documentation: _____